Cambridge (CIE) IGCSE Computer Science

Automated Systems

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What is an automated system?

- An automated system is a computer system that **collects information** and can **react** and **perform tasks** based on the data
- Automated systems are made using **three** components:
  - **Sensors**
  - **Microprocessors**
  - **Actuators**
- A sensor **collects information** and provides the microprocessor with it as an input
- A microprocessor **processes the information** and **makes decisions** based on pre-programmed rules
- An actuator **makes physical changes** based on instructions given by the microprocessor (**outputs**)
- Examples of physical changes include:
  - **Opening/closing a door/valve**
  - **Activating an alarm**

Advantages & Disadvantages of Automated Systems

- There are general advantages and disadvantages of automated systems

Advantages	Disadvantages
Cost - long-term cost saving	**Cost** - short-term expensive to set up
Safer - timely interventions	**Testing** - significant testing must be done before being used
Safer - keeps humans away from hazardous environments	**Security** - open to cyber attacks
Efficient - materials and resources	**Flexibility** - will only react to programmed scenarios

Consistent - results are repeatable	**Maintenance** - needs to be well maintained
	Unethical - can result in major job losses

- There are also scenarios where specific advantages and disadvantages are important such as:

Scenario	Advantages	Disadvantages
Industry	▪ Increased production ▪ Improved quality control	▪ High investment costs ▪ Job losses
Transport	▪ Less accidents & traffic congestion ▪ Increased logistical efficiency	▪ Transportation drivers lose jobs ▪ Technology reliability
Agriculture	▪ Less manual labour ▪ Increased crop yield & resource management	▪ High start-up costs for equipment and sensors
Weather	▪ Accurate forecasting ▪ Improved early warning systems	▪ Dependence on accurate sensor data
Gaming	▪ Personalised experience ▪ AI enhanced opponents add challenge	▪ Reduced creativity & problem solving for players ▪ Repetitive gameplay
Lighting	▪ Increased energy efficiency ▪ Better security with motion-activated systems	▪ Lack of human control with light adjustments
Science	▪ Faster & more precise ▪ Reduced risk of human error	▪ High cost development and maintenance ▪ Potential for algorithm bias

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Worked Example

A theme park has a game where a player tries to run from the start to the finish without getting wet.

The system for the game uses sensors and a microprocessor to spray water at a player as they run past each sensor.

Describe how the sensors and the microprocessor are used in this system

[8]

Answer

- A motion sensor collects data [1]
- This data is converted to digital using ADC [1]
- The sensor sends data to the microprocessor... [1]
- ... where the data is compared to stored data [1]
- If the value is outside range, water will be sprayed [1]
- A signal is sent to the actuator to spray water [1]
- If the value is within range no action is taken [1]
- This runs in a continuous loop [1]